

Heap

This document will be compiled incrementally.

Almost Complete Binary Tree

A **binary tree** ("binary" means every node can have at most two children) is said to be an **almost complete binary tree** when:

- **All levels except the last are completely filled**

Every level of the tree (from the root to the level just before the last) has the maximum number of nodes possible for that level.

- **The last level is filled from left to right**

Nodes at the last level are arranged as far to the left as possible, with no gaps between them.

Why this matters

The **heap data structure** is internally maintained as an **almost complete binary tree**. This structure:

- Guarantees a compact shape
- Allows efficient array-based representation
- Enables $O(\log n)$ insertion and deletion operations

Heap and Priority Queue

A **heap** is most often used in an abstracted form called a **Priority Queue**, which is a very important data structure.

The key advantage of a priority queue is that we can always access the **highest-priority** or **lowest-priority** element efficiently.

When we need to process elements based on a specific priority (for example, highest value, lowest value, earliest deadline, etc.), we construct a heap using that **priority-defining property as the key**.

Operations and Time Complexity

- **Insert:** $O(\log n)$
- **Lookup (peek min / max):** $O(1)$
- **Deletion (remove min / max):** $O(\log n)$

Why this is powerful

This combination of fast access and efficient updates makes heaps / priority queues ideal for:

- Scheduling problems
- Dijkstra's and Prim's algorithms
- Top-K problems
- Real-time systems where priority matters

Heap Property

The **heap property** defines the relationship between a parent node and its child nodes in a heap. This property ensures that the heap remains ordered in a way that supports efficient operations.

The exact relationship depends on the **type of heap** being used. There are two main types:

Min-Heap

- The value of each **parent node is less than or equal to** the values of its child nodes.
- As a result, the **smallest element is always at the root** of the heap.

Max-Heap

- The value of each **parent node is greater than or equal to** the values of its child nodes.
- As a result, the **largest element is always at the root** of the heap.

Key takeaway

The heap property does **not** impose a global ordering like a BST. It only guarantees ordering **between parent and children**, which is sufficient to efficiently retrieve the minimum or maximum element.

Key Points about the Heap Property

- **Local Relationship**

The heap property only enforces ordering between a parent node and its immediate children. It does **not** impose any ordering between sibling nodes or across different levels of the tree.

- **Preservation During Operations**

During insertion or deletion, the heap property may be temporarily violated. A process called **heapification** restores the property by repositioning elements until the heap property is satisfied again.

Array Representation of Heap

Heaps are efficiently represented using an **array**, where the **root node is stored at index 0**. Because a heap is an almost complete binary tree, this array representation is compact and does not waste space.

The relationship between parent and child nodes is defined using simple mathematical formulas:

- **Left child of node at index i :**

$$2 * i + 1$$

- **Right child of node at index i :**

$$2 * i + 2$$

- **Parent of node at index i :**

$$(i - 1) / 2 \text{ (integer division)}$$

Why this works

Because the heap is an almost complete binary tree, nodes are filled level by level from left to right. This predictable structure allows us to compute parent-child relationships directly from indices without using pointers.

Why Array is Preferred Over Pointer-Based Tree Structure

Memory Efficiency

An array-based representation does not need to store explicit pointers to the left child, right child, or parent.

In a pointer-based tree, each node typically requires **three extra references** (left, right, parent). In contrast, an array computes these relationships mathematically using indices, resulting in **significant memory savings**, especially for large heaps.

Cache Locality

Arrays store elements **contiguously in memory**, which leads to much better cache performance.

- When accessing a parent node, its children are usually located nearby in memory.
- This reduces cache misses and improves real-world performance.

In pointer-based trees, nodes may be scattered across memory, leading to frequent cache misses.

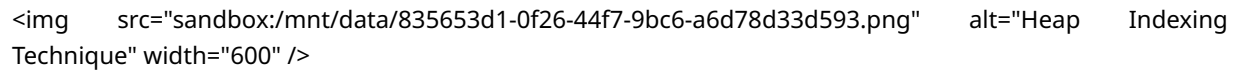
Simplicity

- No pointer management is required during insertion or deletion.
- The heap structure is preserved by simply appending/removing elements at the end of the array and applying **heapification**.
- This makes implementations simpler, safer, and less error-prone.

Bottom line

Heaps are about **performance and predictability**, and array-based implementations win decisively over pointer-based trees in both aspects.

Why This Relationship Is Always True

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Visual representation of heap indexing: for a node at index i , left child is $2i+1$, right child is $2i+2$, and parent is $(i-1)/2$ (integer division).

The index-based parent-child relationship in a heap works because an **almost complete binary tree** is filled **level by level, from left to right**, without gaps.

Level-wise indexing

When nodes are numbered sequentially starting from 0 :

- **Level 0 (root):** 1 node

Index: 0

- **Level 1:** 2 nodes

Indices: $1, 2$

- **Level 2:** 4 nodes

Indices: $3, 4, 5, 6$

- **Level 3:** 8 nodes

Indices: $7, 8, 9, 10, 11, 12, 13, 14$

Each level contains **exactly twice** as many nodes as the previous one.

Parent-child index relationship

For any node at index i :

- Its **left child** is placed at the first available position of the next level segment:

$$2 * i + 1$$

- Its **right child** is placed immediately after that:

$$2 * i + 2$$

This happens deterministically because:

- Nodes are filled left to right
- There are no gaps in the array
- Each level is contiguous in memory

Core reason

This relationship is a direct consequence of:

- The **binary tree structure** (each node has at most two children)
- The **almost complete property** (no missing nodes except possibly at the far right of the last level)

Together, these guarantees make the array index formulas **always valid** for heaps.
